

A hierarchical structure of L-cysteine/Ag NPs/hydrogel for conductive cotton fabrics with high stability against mechanical deformation

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Abstract Electrically conductive fabrics have been increasingly attracting interest due to their assembly facility with wearable devices. Herein, we propose a simple strategy for fabricating flexible, anti-fatigue, and conductive cotton fabrics. Our approach is innovative in fabricating hierarchical coating structure, which is composed of L-cysteine binder, silver nanoparticles (Ag NPs), and a conductive hydrogel coating. By coordinate bonds between the thiol groups and Ag atoms, the cysteine binder effectively enhances the affinity of cotton fibers for the Ag NPs. In-situ deposition of Ag NPs onto the L-cysteine

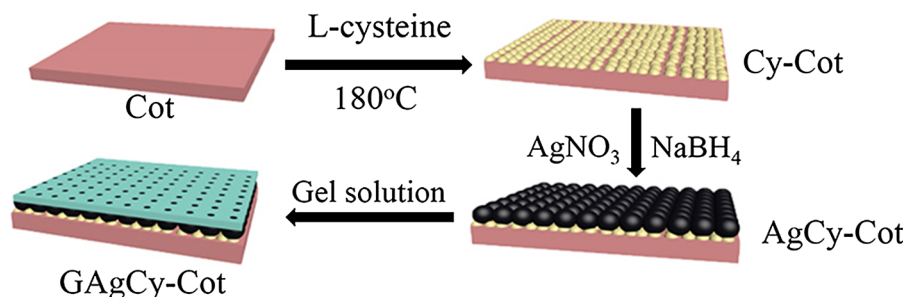
modified cotton fabrics (Cy-Cot) yields electrically conductive fabrics (AgCy-Cot), which have conductivity as high as approximately 10 Ω /sq. Importantly, the top coating of the double-networked, self-healable, and conductive hydrogel provides remarkable electrical stability to the finished cotton fabric (GAgCy-Cot) under stretching, bending, and folding deformation. We believe that the methodology proposed here to fabricate conductive cotton fabric having high durability has potential for smart textile.

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Graphical abstract



Keywords Ag NPs · Conductive · Cotton fabric · Stability

Introduction

Electronic textiles are a major focus of recent efforts in material science and textile engineering, with potential applications ranging from health care (Bandodkar et al. 2016; Chen et al. 2015; Gao et al. 2016; Majumder et al. 2017; Rim et al. 2016; Xiao et al. 2018a, b) to communication (Hussain et al. 2015) and energy storage (Fan et al. 2016; Liu et al. 2017a, b; Wu et al. 2016). In the context of the progress of internet technology and miniaturization of electronic devices, electronic textiles are considered as promising and achievable targets for next generation electronics (Kim et al. 2017; Zeng et al. 2014; Ghosh et al. 2018a, b). A critical component in the electronic textiles is electrically conductive fabrics because they not only act as fundamental conductors to connect other parts like batteries, energy storage devices, and sensors, but also be incorporated into textiles by simple textile manufacturing technologies like tailoring or sewing (Wang et al. 2017). Besides, conductive fabrics themselves are a class of functional materials for various applications, especially in electric heating, electrostatic dissipation, electromagnetic wave absorption, and medical electrodes (Liu et al. 2016; Yetisen et al. 2016; Ghosh et al. 2018a, b; Mondal et al. 2017). After years of experimental and theoretical research, conductive fabric is now reasonably developed, and a number of fabricating strategies of conductive fabric have been reported. Technologies like spinning yarns by blending metal fibers or embroidering conductive yarns to common fabrics

have been the primary methodologies to fabricate conductive fabrics (Ghahremani Honarvar and Latifi 2016; Lai et al. 2017; Stoppa and Chiolerio 2014; Xiao et al. 2018a, b). Unfortunately, these techniques encountered a series of practical problems such as insufficient wearing comfortability, inflexibility and unstable conductivity against deformation. More recently, various coating strategies including printing (Guo et al. 2016; Saleem and Leandro 2017), surface micro-dissolution (Li et al. 2016), dipping and drying (Doga et al. 2016), and deposition (Cheng et al. 2017; Huang et al. 2016; Zhang et al. 2017a, b) have been investigated using conjugate polymer (Hebeish et al. 2016; Liu et al. 2016; Lu et al. 2016), metal (Cheng et al. 2015; Ma et al. 2015; Nechyporchuk et al. 2017; Zhu et al. 2017; He et al. 2018), and carbon nanomaterial (Cataldi et al. 2017; Di et al. 2016; Hu et al. 2010; Jeon et al. 2017; Molina 2016; Zhang et al. 2017a, b) as the conductive coating material. Although these approaches have allowed conductive fabrics to be obtained with improved flexibility and stretchability, few methods can provide modified fabrics with durable conductivity and approving fatigue resistance (Zeng et al. 2014). The main reason is the flaking off of the conducting layer caused by large and frequent deformation. Thus, fabrication of conductive fabrics having high fatigue durability via a scalable and cost-effective process remain as substantial challenges.

In the present work, a unique design of a conductive coating over cotton fabric substrate to improve the electrical stability is proposed. The coating has a hierarchical structure composed of an interlayer of silver nanoparticles (Ag NPs), a top layer of conductive hydrogel, and a bottom layer of L-cysteine binder (Fig. 1a). The L-cysteine layer is designed to enhance adsorptive interaction between the Ag NPs and the surface of cotton fibers. Previous work (Kwak et al.

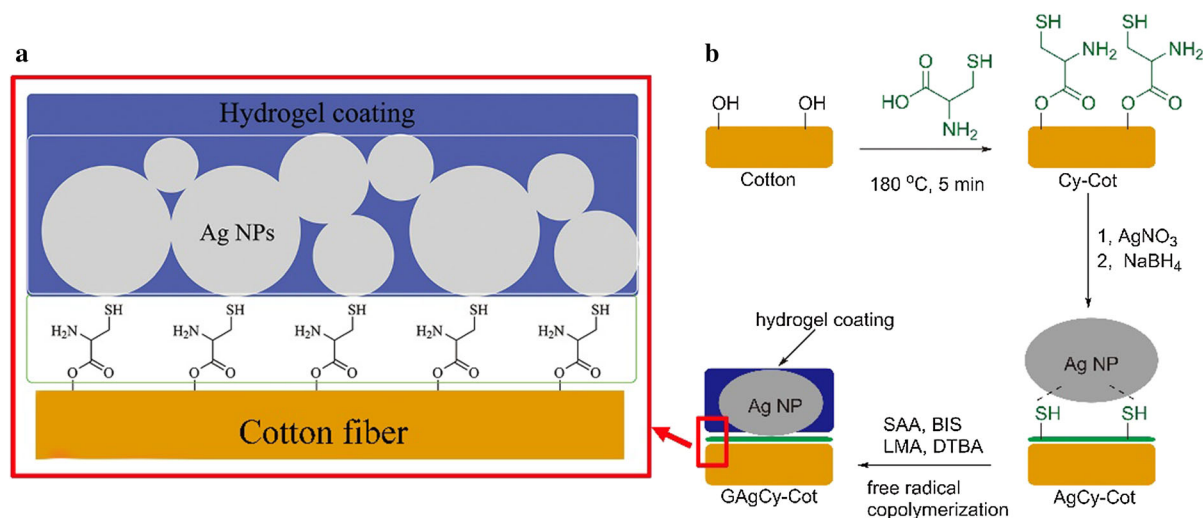


Fig. 1 Schematic illustration (a), and b fabrication process of the hierarchical coating structure on cotton fiber

2015) has demonstrated that amino and thiol groups linked on the cotton fiber surface are very effective for immobilization of Ag NPs because of the strong affinity of the thiol groups to the Ag NPs. The top layer of the composite coating is composed of soft and conductive hydrogel, which is designed to bridge the Ag NPs and to provide robust mechanical and electrical durability against various deformations. We synthesize a poly (acrylic acid) based hydrogel by building double cross-linking networks (Chen et al. 2015; Matsuda et al. 2016), one is obtained by using cross-linking monomer, and another is constructed by the formation of emulsion particles together surfactant molecule. With the hierarchical structured coating, we are able to readily fabricate conductive fabrics that are soft, electrically stable, and durable against deformation. This new coating strategy has potential as a scalable method to impact textile with practical conductive function.

Experimental

Materials

The cotton fabrics were purchased from Suzhou Ke Chuan Textile Co., Ltd. (0.28 mm thickness, 108.5 g/m²). Before used, the cotton samples were cleaned by ultrasonic washing in 2% sodium laurylsulfonate solution (30 min) and ethanol (1 h) and deionized

water (15 min × 3 times) to remove impurities. Paratoluene sulphonic acid, Acrylamide (AM), acrylic acid (AA), sodium hydroxide, N,N'-methylene-bis-acrylamide (BIS), lauryl methacrylate (LMA), dodecyltrimethyl ammonium bromide (DTBA), ammonium persulfate, sodium sulfite, L-cysteine and sodium borohydride were all purchased from Aladdin Co., Ltd (Shanghai, China) and used without further purification. Silver nitrate was purchased from Changzhou Guoyu Co. Ltd (Changzhou, China).

Preparation of conductive hydrogel having doubled networking structure

The hydrogel samples were prepared via a free radical copolymerization of AM, LMA, AA, and BIS in an aqueous solution in the presence of DTBA. Typically, AA monomer (18.0 g, 0.25 mol) was dissolved into distilled water (41.5 mL), neutralized using a solution of NaOH to acquire solution of sodium acrylate (SAA). Then the SAA solution (0.45 mL, 1.52 mol/L) was further mixed with AM (0.75 g, 10.5 mmol), LMA (0.13 g, 0.52 mmol), DTBA (0.48 g, 1.56 mmol), BIS (0.01 g, 0.07 mmol), and water (2.30 mL), and stirred for 10 min using a magnetic stirrer in 150 rpm. The mixture was mixed with ammonium persulfate (120 μL, 0.87 mol/L) and sodium sulfite (120 μL, 1.59 mol/L), further stirred for 10 s, transferred to a mold, and polymerized at

25 °C for 3 h to obtain conductive hydrogel with self-healing ability and flexibility.

Preparation of cotton fabrics having L-cysteine and Ag NPs layers (AgCy-Cot)

A cotton fabric sample (4 cm × 4 cm) was immersed into L-cysteine solution (15.0 mL, 0.99 mol/L) for 4 min, squeezed to weight up 100%, heated at 180 °C for 5 min, rinsed with deionized water (shaken in 100 mL in deionized for 2 min in 35 rpm and rinsed. The washing process was repeated for 3 times), and dried at 80 °C for 2 h to obtain fabric (Cy-Cot) having L-cysteine layer. Then the resulting fabric cotton (Cy-Cot) was immersed into a silver nitrate solution (10.0 mL, 0.89 mol/L) for 5 min, dried at 80 °C for 2 h, immersed into a solution of sodium borohydride solution (20 mL, 0.3 mol/L) for 3 min, rinsed with deionized water (100.0 mL for 3 times), and dried at 80 °C for 1.5 h to obtain fabric (AgCy-Cot) having L-cysteine and Ag NPs layers.

Preparation of gel coated conductive cotton fabrics (GAgCy-Cot)

A polymerizable mixture (3.8 mL) was prepared as mentioned above (“[Preparation of conductive hydrogel having doubled networking structure](#)” section). AgCy-Cot fabric was immersed into the polymerizable mixture for 30 s, squeezed to weight up $190 \pm 3\%$, transferred into a sealed bag at 25 °C, and let it quiescently stand for 3 h to obtain the finished conductive cotton fabric (GAgCy-Cot).

Characterization

The morphology of the cotton fabrics was investigated using a field emission scanning electron microscope set (FE-SEM) set (Ultra-55, Zeiss, Germany) with an energy dispersive x-ray spectrometer (EDS). Fourier transform infrared (FTIR) measurements were performed on a Nicolet Avatar 370 spectrometer (Nicolet Company, Madison, USA) with an attenuated total reflection (ATR) accessory. X-ray photoelectron spectroscopy (XPS) analysis was performed by an AXIS multifunctional X-ray photoelectron spectrometer (Ultra Dld, Shimadzu Ltd., Japan) at a power of 450 W. Thermal Gravity Analysis (TGA) was performed on a thermogravimetric analyzer (TGA/DSC-

1, Mettler-Toledo corp., Switzerland) at a heating rate of 10 °C/min from 30 to 800 °C with nitrogen atmosphere. X-ray diffraction (XRD) analysis was carried out using an X-ray diffractometer (Bruker AXS, Germany). Elemental analysis of silver was performed on an inductively coupled plasma mass spectrometry (ICP-MS, Aglient 7500a, USA). Surface electrical resistance was measured according to the AATCC-76-2005 method. Elongation test of conductive cotton fabric was carried out using a universal material testing machine (Heson HS300C, Shanghai Hesheng Instrument Technology Co., Ltd, China). Fatigue test was performed on a bending machine (Wenzhou, Shangdu Co., Ltd., China) using repeated bending processes (the round-trip distance was 2 cm for the fabrics).

Results and discussion

Figure 1b schematically illustrates the fabrication of the conductive coating on cotton fabric using dipping method. To synthesize the bottom layer, a cotton fabric was dipped in an L-cysteine aqueous solution, squeezed to weight up to 100%, and heated at 180 °C for 5 min to promote the esterification reaction. Then, Ag NPs layer is produced by successive immersions of the resulting fabric (Cy-Cot) in aqueous solutions of AgNO₃ and its reductant, NaBH₄. Finally, dipping the AgCy-Cot fabric into an aqueous mixture composed of acrylamide (AM), sodium acrylate (SAA), N,N'-methylene-bis-acrylamide (BIS), lauryl methacrylate (LMA), and dodecyltrimethyl ammonium bromide (DTBA) (detailed in SI part, Table S1), and following with a free radical copolymerization at 25 °C for 3 h to form conductive hydrogel coating. After be frozen by liquid nitrogen and dried, the SEM cross section of AgCy-Cot cotton coated by gel was shown in Fig. S1.

The chemical modification process was monitored by ATR-FTIR and XPS techniques. Figure 2 compares ATR-FTIR spectrum of Cy-Cot with that of original cotton fabric. The new peak appeared at 1745 cm⁻¹ is assignable to the asymmetric stretching of carboxylic anions, COO⁻, suggesting the linkage of L-cysteine to cellulose chain has formed on fabric surface. As expected, the peak disappeared again after loading Ag NPs, and the reason is ascribed to the Ag coating that covered on the L-cysteine modified surface.

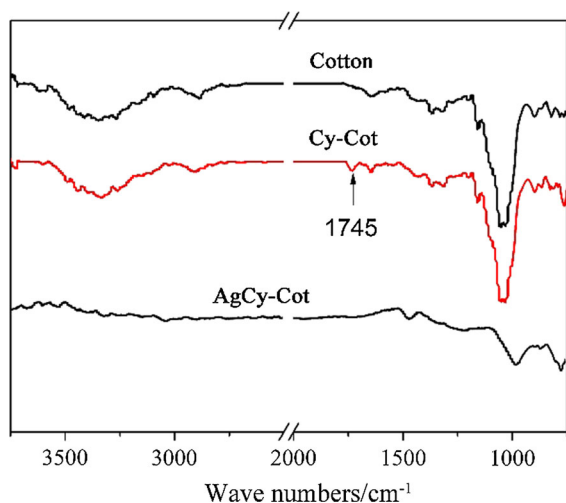


Fig. 2 ATR-FTIR spectra of the original cotton, Cy-Cot cotton and AgCy-Cot cotton

The modified cotton fabric was further assessed by TGA tests in a N_2 flow. TGA (Fig. 3a) and DTG curves (Fig. S2) suggest that the cotton fabrics degrade via three stages: evaporation of absorbed water (for the temperature range up to 290 °C), breakage of the ether bonds (300–450 °C), and carbonization to form carbon skeleton (above 450 °C). It was surmised that the slight differences in the TGA profiles are caused by the catalytic action of the Ag NPs in the decomposing reactions. The content of the Ag NPs was estimated using ICP-MS analysis technology (Fig. 3b). Basing on the TGA data and ICP-MS analysis, it was found that L-cysteine concentration is an effective factor to influence loading of Ag NPs. The concentration of L-cysteine used for dipping cotton fabric was decided by considering loading quantity of the Ag NPs and conductivity. As shown in Fig. 3c, d, with higher concentration, more silver loading quantities can be obtained, but the surface resistance becomes almost constant with further increase of the L-cysteine concentration. Based on our experimental results, 0.07 g/mL concentration provides sufficient conductivity.

XPS spectroscopy was employed to better understand the interfacial chemical composition of the modified samples. As shown in Fig. 4a, AgCy-Cot fabric shows additional N1s (399.0 eV), S2p (162.1 eV) and Ag3d (368.0 eV) signals when compared with original cotton fabric. XPS survey spectrum of original cotton present only two peaks at

285 eV and 533 eV, which are assigned to C1s and O1s respectively. The C1s peak decomposition of origin cotton reveals three major components at 284.6, 286.7, and 288.2 eV assigned to C–C/C–H (hydrocarbon), C–O and C–O–C bonds, respectively. Moreover, the XPS spectra in high resolution (Fig. 4b, c) indicate that the C1s peak of the Cy-Cot sample changed to more complex after the two modification steps, i.e., new peaks at 287.1 eV (C–S) and 288.6 eV (C=O) and 285.5 eV (C–N) are presented. These XPS results suggest again that L-cysteine have linked on the fiber surface by ester groups (Meng et al. 2016; Xu et al. 2017; Zhang et al. 2018). Interestingly, XPS spectroscopy study of S2p (Fig. 4d) reveals that there is a new peak (161.3 eV) on the high resolution spectrum of AgCy-Cot fabric by comparing with the Cy-Cot sample before loading Ag NPs. This can be attributed to the highly de-electronated state of S elements (Porcaro et al. 2016). The amino groups of Cys show affinity with silver atoms as well, as shown in Fig. S3, the N1s XPS spectrum of Cy-Cot fabric has a splitting peak at 405.1 eV, which is attributable to the highly de-electronated N elements. The results provide an irrefutable evidence for the coordination bonds between the amino and thiol groups of the cysteine moieties and the Ag NPs.

The enhanced affinity of the L-cysteine moieties for the Ag NPs increases the quantity of Ag NPs, leading to the lower surface resistances closed to 10 Ω /sq. These resistances are lower than those of several previously reported conductive fabrics, such as conductive fabrics based on reduced graphene oxide (600 Ω /sq) (Ren et al. 2017), nanosilver based fabrics fabricated using layer by layer method (10×10^4 Ω /sq) (Ahmed and Emam 2017), and conductive polymer based fabrics (15 Ω /sq) (Stempien et al. 2015). They are even similar to those of recently reported Ag nanowire-based fabrics (10 Ω /sq) (Wei et al. 2016). However, the conductive fabrics in the present work are very weak at capability for enduring mechanical deformation.

To improve the anti-deformability, a hydrogel with appropriate mechanical property and conductivity was designed as a protective coating on the Ag NPs layer. Conductive hydrogels have possessed widespread application potentials in many fields. Conductive hydrogels could be sorted several types, such as conductive polymers-based conductive hydrogel (Du et al. 2011; Guo et al. 2015; Lu et al. 2014) and

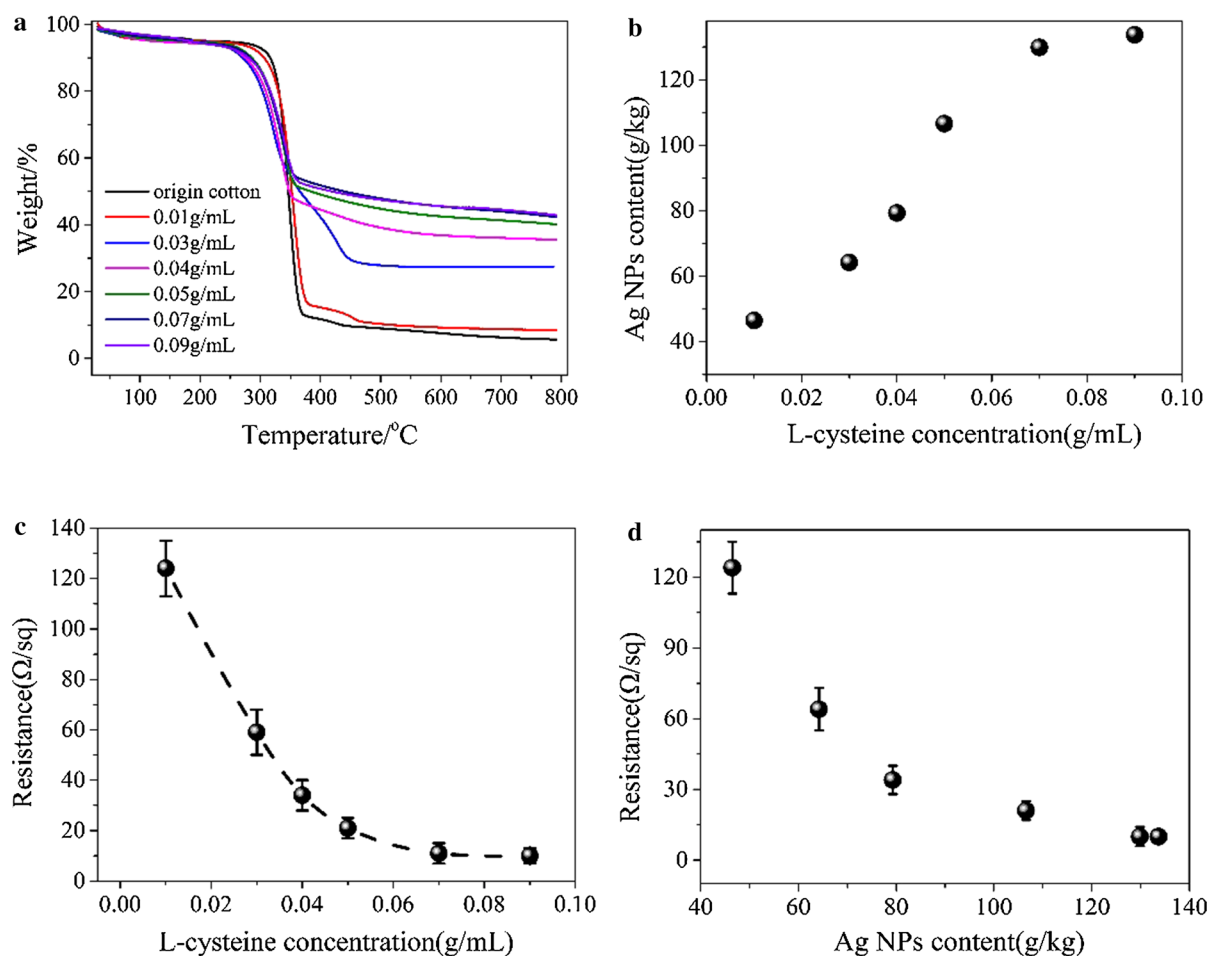


Fig. 3 **a** TG curves, **b** Ag NPs content with different L-cysteine concentration, **c** surface resistances of the AgCy-Cot fabrics fabricated using a variety of L-cysteine concentrations, **d** Ag NPs content effects on surface resistance

polyelectrolyte conductive hydrogel (Tsai et al. 2011). A poly (acrylic acid) based hydrogel was selected and the double-networking strategy developed by Gong and co-workers (Haque et al. 2012; Sun et al. 2013; Webbe et al. 2007) was chosen to remedy the brittle nature of general poly (acrylic acid) hydrogels. As schematically shown in Fig. S4, besides the polymerizable cross-linker, BIS monomer, numerous latex particles that coagulated via hydrophobic association of the LMA moieties and DTBA surfactant were used as the second hydrophobic cross-linking centers. The first cross-linker forms a hydrophilic network to provide the hydrogel with a soft mechanical property, and the second makes the hydrogel dynamically self-healable (Liu et al. 2017a, b). The hydrogels displayed good mechanical flexibility, e.g., they are able to be stretched to higher elongation than 1500%, and even

bent to a knot (Fig. 5a). Figure 5a shows the representative optical images of how the hydrogel was healed after being completely cut. After the two fractured surfaces contacted again, they were totally restored after 24 h at room temperature. Obviously, the approvable resistivity is corresponding to the polyelectrolyte structure. Before applying this hydrogel onto the fabric samples, the polymerization conditions were first optimized (Table S1). Since the BIS content affects the cross-linked density of the hydrogel, the resistivity of double-networking gel decreased firstly and then increased with increasing BIS, and the lowest point reached $60.5 \Omega/\text{cm}^{-1}$ when BIS mole fraction was 0.5% (Fig. 5b). DTBA also plays an important role in the double-networking gel system because it participates in forming micelles with the LMA moieties. Compared with the chemical

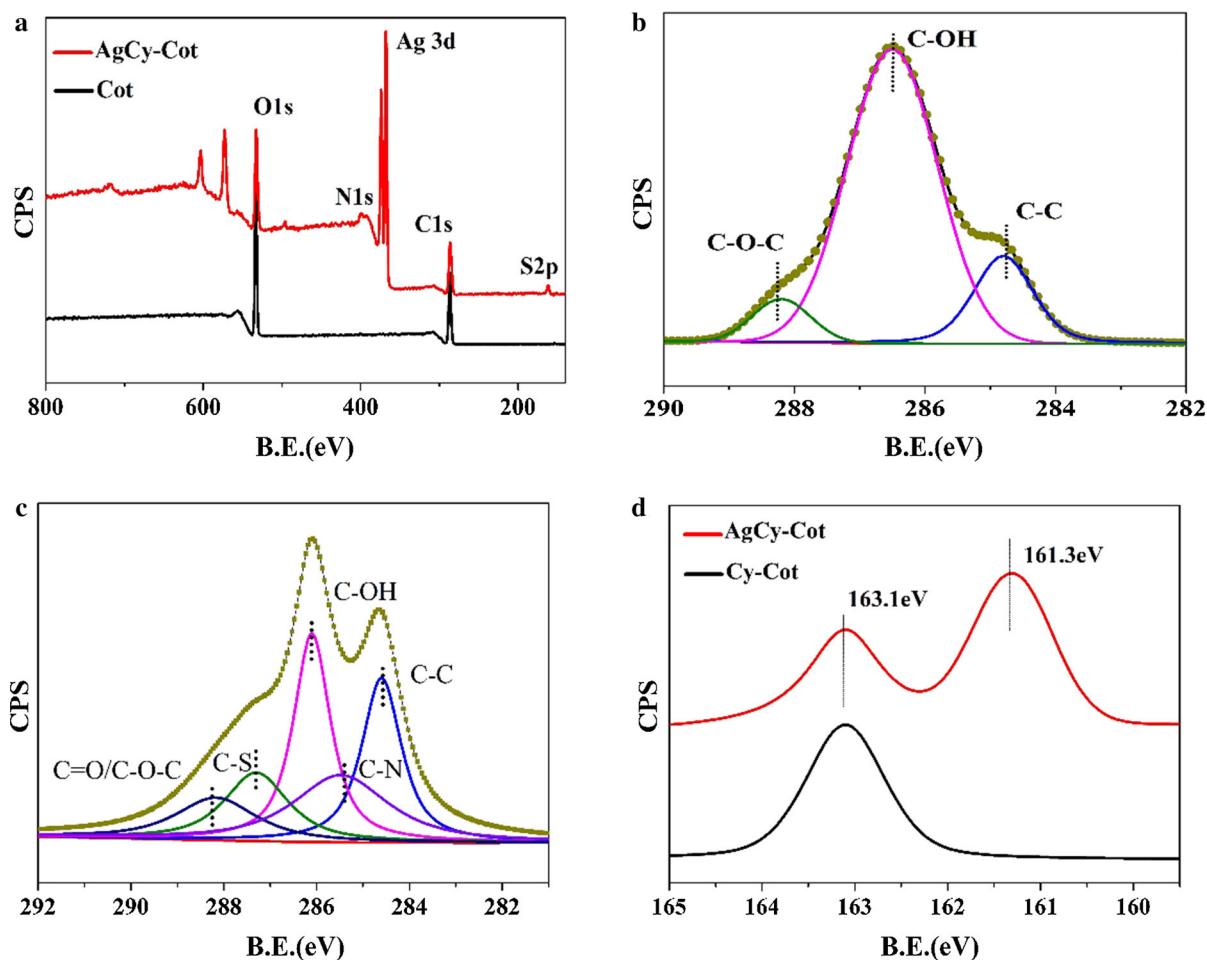


Fig. 4 **a** XPS survey spectra of original cotton and AgCy-Cot fabrics, **b** C1s XPS spectra of original cotton fabric, **c** C1s XPS spectra of Cy-Cot fabric, **d** S2p XPS spectra of Cy-Cot and AgCy-Cot fabrics

crosslinking, these micelles are hydrophobic and have a bigger volume, thus the resistivity of the hydrogel increases almost linearly (Fig. 5c). As the important component in the physical cross-linking structure, DTBA influences elongation and breaking strength of the gel as well (Table S1). Taking various factors into consideration, 0.67 mol/L DTBA, 0.03 mol/L BIS, 4.56 mol/L AM, 1.52 mol/L SAA, 0.23 mol/L LMA, and 120 μ L initiator, was selected as the optimized concentration for the copolymerization to prepare the conductive hydrogel.

On the basis of the optimization work, the soft, stretchable, and conductive hydrogel was applied to the cotton fabric surface. After mixing the monomers (SAA, AM, LMA, and BIS), surfactant (DTBA), and water, a well emulsified emulsion could be obtained.

Then, the Ag loaded cotton fabric, AgCy-Cot, was dipped into the polymerizable emulsion at 25 $^{\circ}$ C, allowing the in situ copolymerization on the surface of the cotton fabric. This treatment results in a conductive hydrogel coating on the cotton surface.

Figure 6a, b compares the optical images of the original cotton fabric and the modified cotton fabric (GAgCy-Cot) with the conductive hierarchical coating. Cotton fabric appears in white before the modifications, but shows a black colour which indicated that Ag NPs were deposited on the cotton fabric successfully. The finished fabric, GAgCy-Cot, is bendable and twistable as untreated cotton fabric, meaning that there are not serious damages on cotton flexibility.

As shown in Fig. 7a, b, SEM image shows that the surface of pristine cotton fibers is flat and smooth. But

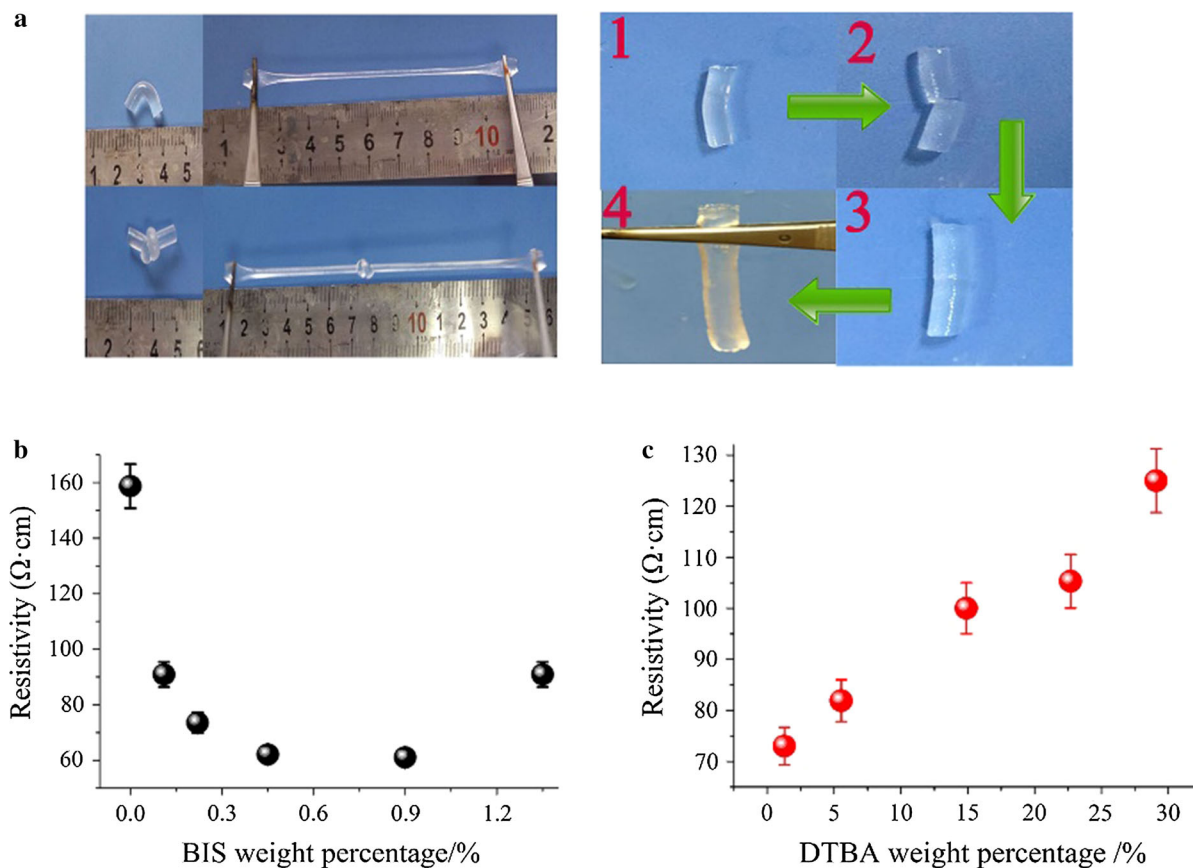


Fig. 5 **a** Optical images of the hydrogel in various deformation conditions and sequential optical images of the hydrogel's self-healing, content function of BIS (**b**), and DTBA (**c**) on conductivity of the hydrogel

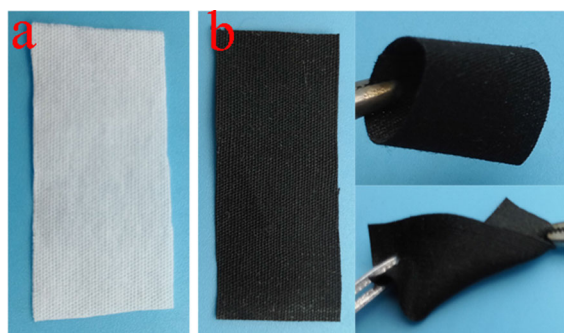


Fig. 6 Optical images of original cotton fabric (**a**), and GAgCy-Cot cotton fabric (**b**)

Fig. 7c, d suggests that the Cy-Cot cotton fabric has rougher surface because of the presence of L-cysteine. After coated by Ag NPs, a lots of nanoparticles were observed (Fig. 7e, f). The surface morphologies of a rough gel layer coated on the fabric surface can be observed from Fig. 7g, h. There are also some gels in

the gap between the fibers. The results show that the gel coated on the cotton successfully. Based on high magnification SEM image of AgCy-Cot (Fig. 8a), size distribution of Ag particles were obtained, which was shown in Fig. 8b. Ag NPs ranged from 10 to 180 nm have an average diameter of 38.9 nm. Fig. S5 compares the XRD patterns of the original and AgCy-Cot cotton fabrics. The characteristic peaks at $2\theta = 38.6^\circ$, 44.7° , 64.9° and 77.8° correspond to (111), (200), (220), and (311) planes of silver crystal (JCPDS cards 4-0783), respectively. Based on Scherrer formula (Montazer and Komeily Nia 2015), the average size of Ag NPs loaded on the cotton fabric was calculated as 41.5 nm. The result is in good agreement with the SEM observation. Elemental mapping images (Fig. 8c, d) of energy dispersive X-ray spectroscopy (EDS) illustrate the distribution of the Ag and S elements, indicating that Ag and S are uniformly deposited on the AgCy-Cot cotton fabric.

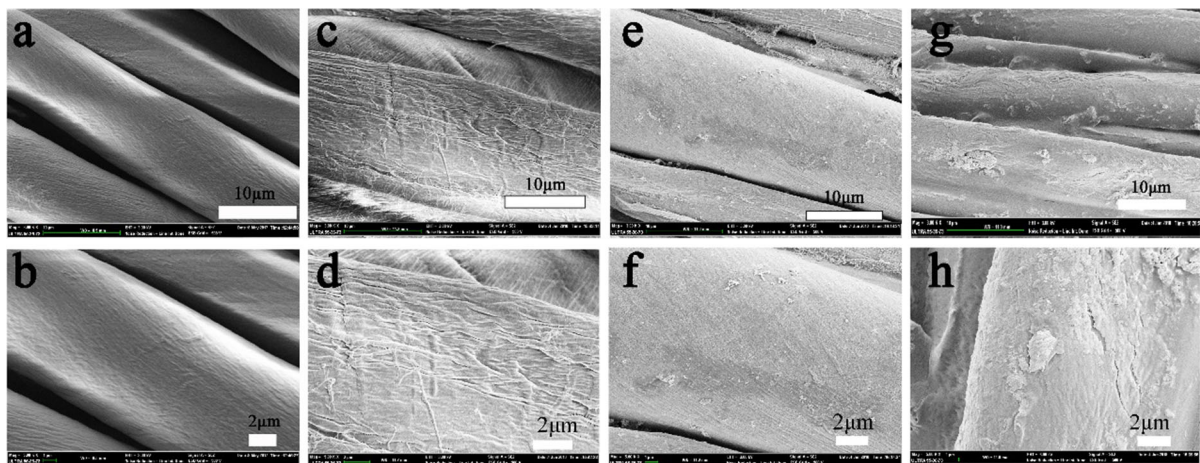


Fig. 7 SEM image of origin cotton (a, b), Cy-Cot cotton (c, d), AgCy-Cot cotton (e, f) and GAgCy-Cot cotton (g, h)

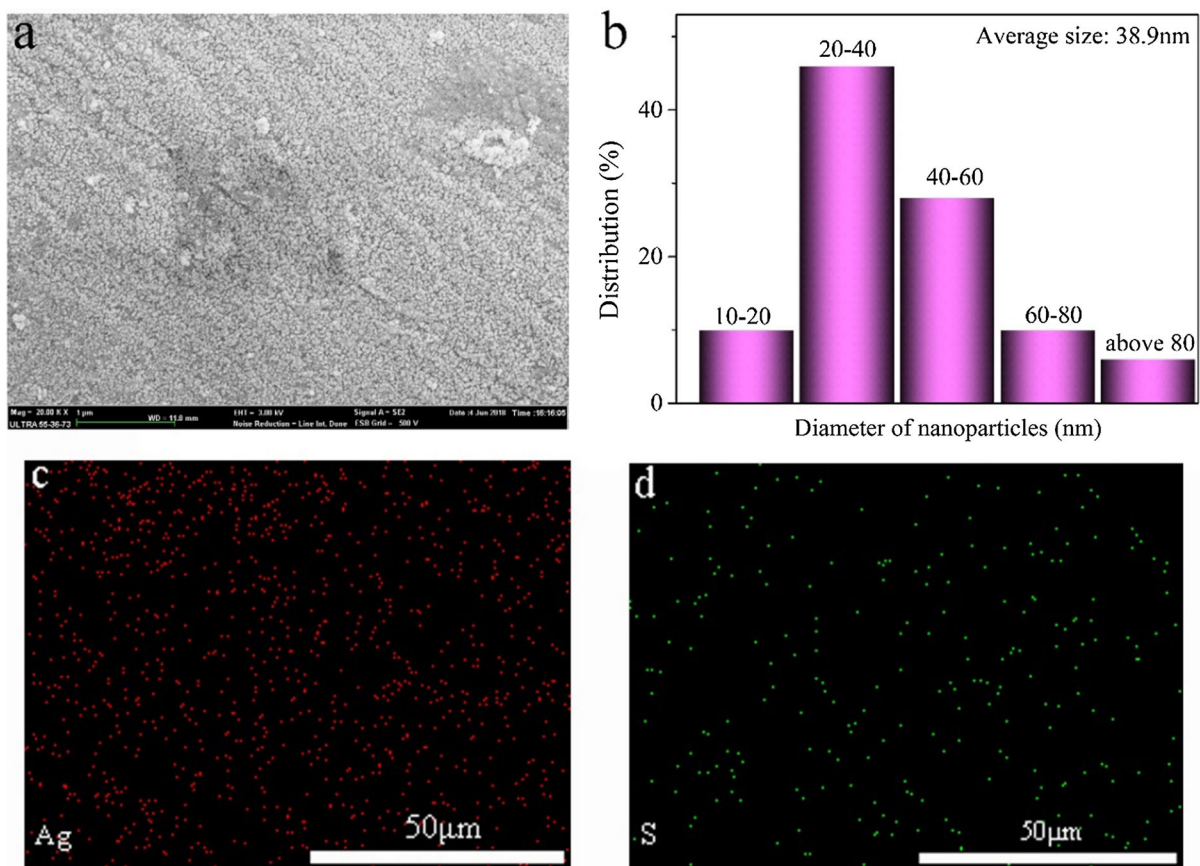


Fig. 8 a High magnification ($\times 20,000$) SEM images, b the size distribution graphs of the AgCy-Cot, mapping of the AgCy-Cot fabric Ag (c) and S (d)

The electrical conductivity of the cotton fabrics was evaluated by determining the surface resistance of the fabric according to AATCC-76-2005 method, as

shown in Fig. 9a. The surface resistance of the different cotton was shown in Table 1. By comparing the Cy + Ag-Cot (4) and Ag-Cot (5), the necessary of

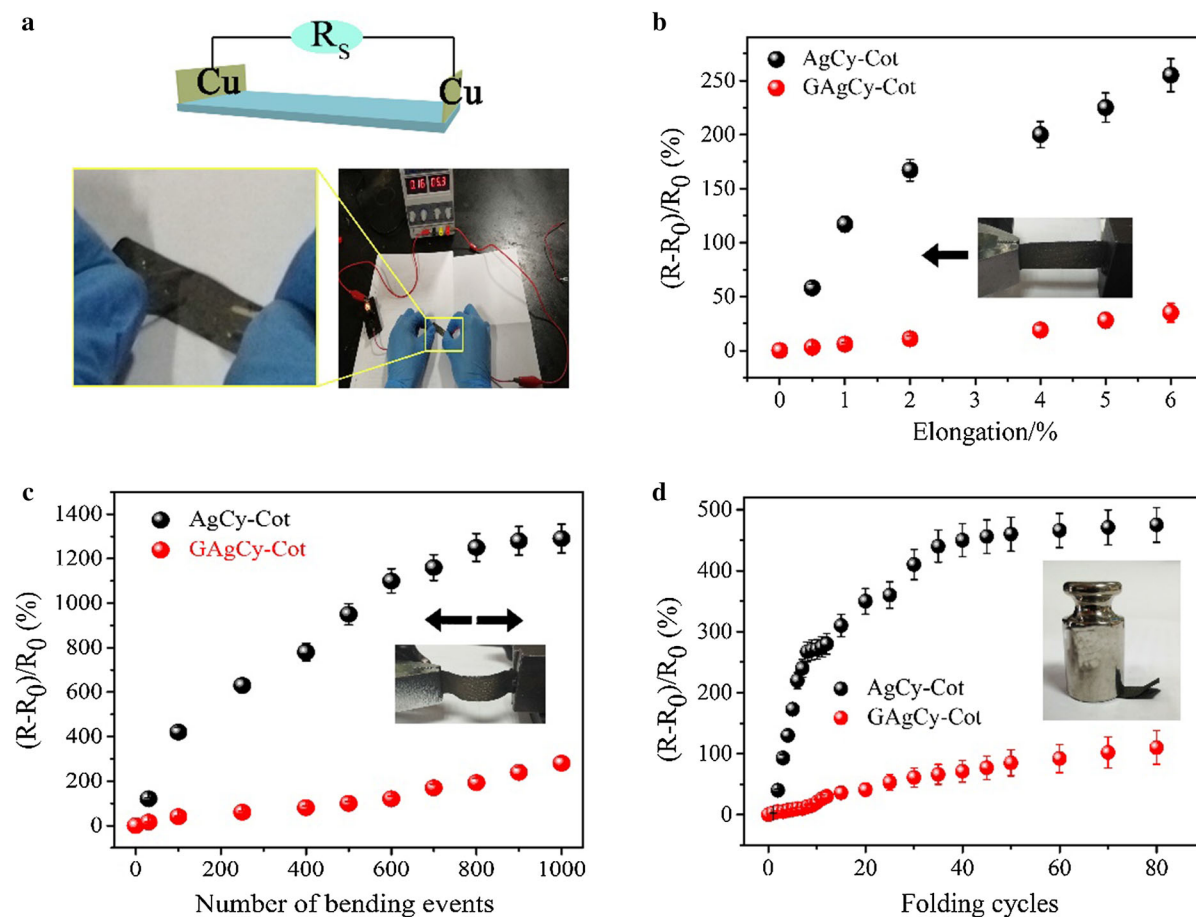


Fig. 9 **a** Schematic illustration of the electrical resistance test and optical images of the conductivity of GAgCy-Cot fabric in a circuit with power device and lamp. Variation in electrical

resistance of GAgCy-Cot fabric with strain (**b**), bending test cycles (**c**), and folding-unfolding test cycles (**d**)

the esterification of L-cysteine is cleared. Without the esterification, very few L-cysteine molecules are remained on the surface after washing, leading to the similar surface resistance to the Ag-Cot (5). The surface resistance of AgCy-Cot fabric is $10 \Omega/\text{sq}$, but the surface resistance of GAgCy-Cot increases to $35 \Omega/\text{sq}$ after coated with conductive hydrogel. This is

because hydrogel has a lower conductivity than silver layer (Table 1).

Interestingly enough, anti-deformation behaviour and anti-fatigue capability of the finished cotton fabric are significantly improved by the coating of the conductive hydrogel layer. As shown in Fig. 9b, we examined the response of GAgCy-Cot in surface

Table 1 surface resistance of the different stage cotton

Fabrics ^a	1 Cot	2 Cy-Cot	3 G-Cot	4 Cy + Ag-Cot	5 Ag-Cot	6 AgCy-Cot	7 GAg-Cot	8 GAgCy-Cot
Resistance (Ω/sq)	∞	∞	$9.32 \times 10^4 \pm 1435$	331 ± 5	350 ± 5	10 ± 1	410 ± 7	35 ± 3

^aFabric 1 (Cot): cotton; 3 (G-Cot): cotton fabric with the hydrogel coating; 4 (Cy + Ag-Cot): the L-cysteine was deposited onto the cotton at room temperature without esterification. 5 (Ag-Cot): cotton fabric with the layer of Ag NPs; 7 (GAg-Cot): Ag-Cot fabric with the hydrogel coating

resistance against mechanical stretch. The elongation rate was set at 1 mm/min, and the change in surface resistance was recorded. It is obvious that AgCy-Cot is very sensitive for stretching, as the doubled surface resistance reached when the strain was stretched to approximately 6%. Whereas, GAgCy-Cot cotton fabric exhibit strong stability against stretching deformation until it was broken. The conductive stability against bending deformation was next evaluated by repeating bending tests. This bending test reflects the anti-bending capability in a simulation to the clothes under wearing conditions. By comparing to AgCy-Cot, the data shown in Fig. 9c exhibit great improvement on the electrical stability after introduced the hydrogel coating. Furthermore, conductive stability against folding was estimated by repeating complete (180°) folding-unfolding tests. The severity of the folding event was ensured by pressing the fold line with a 100 g weight. As shown in Fig. 9d, the surface resistance of AgCy-Cot increases obviously with repeated folding times, but GAgCY-Cot keeps almost constant even after 80 cycles.

Based on comprehensive reasoning and key-factor analysis of these experimental data, we summarized the durability improving mechanism of our designed coating structure in a schematic image as shown in Fig. 1a. The L-Cysteine moieties linked on the cotton fabric surface provide strong adhesion force for the Ag NPs via coordination bond with the Ag atoms on the surface of Ag NPs. They act as a good binder layer to prevent peeling of the silver layer from fiber surface, and increase loading of the Ag NPs. The hydrogel layer plays two roles for enhancing the anti-fatigue capability against large deformation. First, as the continuous phase between the Ag NPs, its semi-conductivity makes it able to bridge the Ag NPs. Second, its soft network structure can adsorb considerable mechanical energy and thus keep the Ag NPs in contact.

Conclusions

In summary, we have demonstrated a novel strategy for the fabrication of flexible, anti-fatigue, and conductive cotton fabrics based on a hierarchical coating composed of L-cysteine binder, layer of Ag NPs, and a conductive hydrogel. The cysteine binder effectively enhances the adhesion force between the Ag NPs and

fiber surface, and the hydrogel layer offers remarkable improvement of the durability of the finish on the cotton fabric. Especially, the hydrogel layer, which was prepared by an in situ copolymerization of SAA and LMA in the presence of DTBA, forms double-networking structure by covalent bonds and latex particles, offering significant contribution to the electrical stability of the Ag NPs based conductive cotton fabrics. With the hierarchically structured coating, we are able to readily fabricate conductive fabrics that are very soft, electrically stable, and durable against deformation. This new coating strategy has potential as a scalable method to impact the design of textiles with practical conductivity.

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Compliance with ethical standards

Conflict of interest There are no conflicts to declare.

References

- Ahmed HB, Emam HE (2017) Layer by layer assembly of nanosilver for high performance cotton fabrics. *Fibers Polym* 17:418–426. <https://doi.org/10.1007/s12221-016-5814-3>
- Bandodkar AJ, Jeerapan I, Wang J (2016) Wearable chemical sensors: present challenges and future prospects. *ACS Sens* 1:464–482. <https://doi.org/10.1021/acssensors.6b00250>
- Cataldi P, Ceseracciu L, Athanassiou A, Bayer IS (2017) Healable Cotton-graphene nanocomposite conductor for wearable electronics. *ACS Appl Mater Interfaces* 9:13825–13830. <https://doi.org/10.1021/acsami.7b02326>
- Chen Q, Zhu L, Chen H, Yan H, Huang L, Yang J, Zheng J (2015) A novel design strategy for fully physically linked double network hydrogels with tough, fatigue resistant, and self-healing properties. *Adv Funct Mater* 25:1598–1607. <https://doi.org/10.1002/adfm.201404357>
- Cheng Y, R-r Wang, Sun J, Gao L (2015) Highly conductive and ultrastretchable electric circuits from covered yarns and silver nanowires. *ACS Nano* 9:3887–3895. <https://doi.org/10.1021/nn5070937>
- Cheng N-y, Zhang L-s, Joon Kim J, Andrew TL (2017) Vapor phase organic chemistry to deposit conjugated polymer films on arbitrary substrates. *J Mater Chem C* 5:5787–5796. <https://doi.org/10.1039/c7tc00293a>
- Di J, Zhang X, Yong Z, Zhang Y, Li D, Li R, Li Q (2016) Carbon-nanotube fibers for wearable devices and smart textiles. *Adv Mater* 28:10529–10538. <https://doi.org/10.1002/adma.201601186>

- Doga D, Sahin C, Sevim Polat G, Husnu Emrah U (2016) Silver nanowire decorated heatable textiles. *Nanotechnology* 27:435201. <https://doi.org/10.1088/0957-4484/27/43/435201>
- Du R, Xu Y, Luo Y, Zhang X, Zhang J (2011) Synthesis of conducting polymer hydrogels with 2D building blocks and their potential-dependent gel-sol transitions. *Chem Commun* 47:6287–6299. <https://doi.org/10.1039/c1cc10915d>
- Fan FR, Tang W, Z-l Wang (2016) Flexible nanogenerators for energy harvesting and self-Powered electronics. *Adv Mater* 28:4283–4305. <https://doi.org/10.1002/adma.201504299>
- Gao W, Emaminejad S, Nyein HYY, Challa S, Chen K, Peck A, Fahad HM, Ota H, Shiraki H, Kiriya D, Lien D-H, Brooks GA, Davis RW, Javey A (2016) Fully integrated wearable sensor arrays for multiplexed in situ perspiration analysis. *Nature* 529:509–514. <https://doi.org/10.1038/nature16521>
- Ghahremani Honarvar M, Latifi M (2016) Overview of wearable electronics and smart textiles. *J Text Inst* 108:631–652. <https://doi.org/10.1080/00405000.2016.1177870>
- Ghosh S, Mondal S, Ganguly S, Remanan S, Singha ND, Das NC (2018a) Carbon nanostructures based mechanically robust conducting cotton fabric for improved electromagnetic interference shielding. *Fibers Polym* 19:1064–1073. <https://doi.org/10.1007/s12221-018-7995-4>
- Ghosh S, Remanan S, Mondal S, Ganguly S, Das P, Singha ND, Das NC (2018b) An approach to prepare mechanically robust full IPN strengthened conductive cotton fabric for high strain tolerant electromagnetic interference shielding. *Chem Eng J* 344:138–154. <https://doi.org/10.1016/j.cej.2018.03.039>
- Guo H-t, He W-n, Lu Y, Zhang X (2015) Self-crosslinked polyaniline hydrogel electrodes for electrochemical energy storage. *Carbon* 92:133–141. <https://doi.org/10.1016/j.carbon.2015.03.062>
- Guo Y, Otley MT, Li M, Zhang X, Sinha SK, Treich GM, Sotzing GA (2016) PEDOT:PSS “Wires” printed on textile for wearable electronics. *ACS Appl Mater Interfaces* 8:26998–27005. <https://doi.org/10.1021/acsami.6b08036>
- Haque MA, Kurokawa T, Gong JP (2012) Super tough double network hydrogels and their application as biomaterials. *Polymers* 53:1805–1822. <https://doi.org/10.1016/j.polymers.2012.03.013>
- He S, Xin B, Chen Z, Liu Y (2018) Flexible and highly conductive Ag/G-coated cotton fabric based on graphene dipping and silver magnetron sputtering. *Cellulose* 25:3691–3701. <https://doi.org/10.1007/s10570-018-1821-4>
- Hebeish A, Farag S, Sharaf S, Shaheen TI (2016) Advancement in conductive cotton fabrics through in situ polymerization of polypyrrole-nanocellulose composites. *Carbohydr Polym* 151:96–102. <https://doi.org/10.1016/j.carbpol.2016.05.054>
- Hu L, Pasta M, Mantia FL, Cui L, Jeong S, Deshazer HD, Choi JW, Han MS, Cui Y (2010) Stretchable, porous, and conductive energy textiles. *Nano Lett* 10:708–714. <https://doi.org/10.1021/nl903949m>
- Huang G-x, Liu L-m, Wang R, Zhang J, Sun X-m, Peng H-s (2016) Smart color-changing textile with high contrast based on a single-sided conductive fabric. *J Mater Chem C* 4:7589–7594. <https://doi.org/10.1039/c6tc02051h>
- Hussain AM, Ghaffar FA, Park SI, Rogers JA, Shamim A, Hussain MM (2015) Metal/Polymer based stretchable antenna for constant frequency Far-Field communication in wearable electronics. *Adv Funct Mater* 25:6565–6575. <https://doi.org/10.1002/adfm.201503277>
- Jeon JW, Cho SY, Jeong YJ, Shin DS, Kim NR, Yun YS, Kim HT, Choi SB, Hong WG, Kim HJ, Jin HJ, Kim BH (2017) Pyroprotein-based electronic textiles with high stability. *Adv Mater* 29:1605479. <https://doi.org/10.1002/adma.201605479>
- Kim J, Kumar R, Bhandodkar AJ, Wang J (2017) Advanced materials for printed wearable electrochemical devices: a review. *Adv Electron Mater* 3:1600260. <https://doi.org/10.1002/aelm.201600260>
- Kwak WG, Oh MH, Gong MS (2015) Preparation of silver-coated cotton fabrics using silver carbamate via thermal reduction and their properties. *Carbohydr Polym* 115:317–324. <https://doi.org/10.1016/j.carbpol.2014.08.070>
- Lai Y-c, Deng J, Zhang S-l, Niu S, Guo H, Z-l Wang (2017) Single-thread-based wearable and highly stretchable triboelectric nanogenerators and their applications in cloth-based self-powered human-interactive and biomedical sensing. *Adv Funct Mater* 27:1604462. <https://doi.org/10.1002/adfm.201604462>
- Li L-q, Fan T, Hu R-m, Liu Y-p, Lu M (2016) Surface micro-dissolution process for embedding carbon nanotubes on cotton fabric as a conductive textile. *Cellulose* 24:1–8. <https://doi.org/10.1007/s10570-016-1160-2>
- Liu S-c, Hu M-j, Yang J (2016) A facile way of fabricating a flexible and conductive cotton fabric. *J Mater Chem C* 4:1320–1325. <https://doi.org/10.1039/c5tc03679h>
- Liu W, M-s Song, Kong B, Cui Y (2017a) Flexible and stretchable energy storage: recent advances and future perspectives. *Adv Mater* 29:1603436. <https://doi.org/10.1002/adma.201603436>
- Liu X, L-j Duan, G-h Gao (2017b) Rapidly self-recoverable and fatigue-resistant hydrogels toughened by chemical crosslinking and hydrophobic association. *Eur Polymer J* 89:185–194. <https://doi.org/10.1016/j.eurpolymj.2017.02.025>
- Lu Y, He W-n, Cao T, Guo H-t, Zhang Y-y, Li Q, Shao Z, Cui Y, Zhang X (2014) Elastic, conductive, polymeric hydrogels and sponges. *Sci Rep* 4:5792–5799. <https://doi.org/10.1038/srep05792>
- Lu M, R-y Xie, Z-l Liu, Zhao Z-y XuH, Z-p Mao (2016) Enhancement in electrical conductive property of polypyrrole-coated cotton fabrics using cationic surfactant. *J Appl Polym Sci* 133:43601–43606. <https://doi.org/10.1002/app.43601>
- Ma R-j, Kang B, Cho S, Choi M, Baik S (2015) Extraordinarily high conductivity of stretchable fibers of polyurethane and silver nanoflowers. *ACS Nano* 9:10876–10886. <https://doi.org/10.1021/acs.nano.5b03864>
- Majumder S, Mondal T, Deen MJ (2017) Wearable sensors for remote health monitoring. *Sensors* 17:130–174. <https://doi.org/10.3390/s17010130>
- Matsuda T, Nakajima T, Fukuda Y, Hong W, Sakai T, Kurokawa T, Chung UI, Gong JP (2016) Yielding criteria of

- double network hydrogels. *Macromolecules* 49:1865–1872. <https://doi.org/10.1021/acs.macromol.5b02592>
- Meng F-d, Yu Y-l, Zhang Y-m, Yu W-j, Gao J-m (2016) Surface chemical composition analysis of heat-treated bamboo. *Appl Surf Sci* 371:383–390. <https://doi.org/10.1016/j.apsusc.2016.03.015>
- Molina J (2016) Graphene-based fabrics and their applications: a review. *RSC Adv* 6:68261–68291. <https://doi.org/10.1039/c6ra12365a>
- Mondal S, Ganguly S, Das P, Bhawal P, Das TK, Nayak L, Khastgir D, Das NC (2017) High-performance carbon nanofiber coated cellulose filter paper for electromagnetic interference shielding. *Cellulose* 24:5117–5131. <https://doi.org/10.1007/s10570-017-1441-4>
- Montazer M, Komeily Nia Z (2015) Conductive nylon fabric through in situ synthesis of nano-silver: preparation and characterization. *Mater Sci Eng C* 56:341–347. <https://doi.org/10.1016/j.msec.2015.06.044>
- Nechyporchuk O, Yu J, Nierstrasz VA, Bordes R (2017) Cellulose nanofibril-based coatings of woven cotton fabrics for improved inkjet printing with a potential in e-textile manufacturing. *ACS Sustain Chem Eng* 5:4793–4801. <https://doi.org/10.1021/acssuschemeng.7b00200>
- Porcaro F, Carlini L, Ugolini A, Visaggio D, Visca P, Fratoddi I, Venditti I, Meneghini C, Simonelli L, Marini C, Olszewski W, Ramanan N, Luisetto I, Battocchio C (2016) Synthesis and structural characterization of silver nanoparticles stabilized with 3-mercaptopropylsulfonate and 1-thiogluco mixed thiols for antibacterial applications. *Materials (Basel)* 9:1028–1042. <https://doi.org/10.3390/ma9121028>
- Ren J-s, Wang C-x, Zhang X, Carey T, Chen K-l, Yin Y, Torrisi F (2017) Environmentally-friendly conductive cotton fabric as flexible strain sensor based on hot press reduced graphene oxide. *Carbon* 111:622–630. <https://doi.org/10.1016/j.carbon.2016.10.045>
- Rim Y-S, Bae SH, Chen H, De Marco N, Yang Y (2016) Recent progress in materials and devices toward printable and flexible sensors. *Adv Mater* 28:4415–4440. <https://doi.org/10.1002/adma.201505118>
- Saleem K, Leandro L (2017) Recent advances of conductive nanocomposites in printed and flexible electronics. *Smart Mater Struct* 26:083001. <https://doi.org/10.1088/1361-665X/aa7373>
- Stempien Z, Rybicki T, Rybicki E, Kozanecki M, Szykowska MI (2015) In-situ deposition of polyaniline and polypyrrole electroconductive layers on textile surfaces by the reactive ink-jet printing technique. *Synth Met* 202:49–62. <https://doi.org/10.1016/j.synthmet.2015.01.027>
- Stoppa M, Chiolerio A (2014) Wearable electronics and smart textiles: a critical review. *Sensors* 14:11957–11992. <https://doi.org/10.3390/s140711957>
- Sun T-l, Kurokawa T, Kuroda S, Ihsan AB, Akasaki T, Sato K, Haque MA, Nakajima T, Gong JP (2013) Physical hydrogels composed of polyampholytes demonstrate high toughness and viscoelasticity. *Nat Mater* 12:932–937. <https://doi.org/10.1038/nmat3713>
- Tsai T-S, Pillay V, Choonara YE, Du Toit LC, Modi G, Naidoo D, Kumar P (2011) A polyvinyl alcohol-polyaniline based electro-conductive hydrogel for controlled stimuli-actuable release of indomethacin. *Polymers* 3:150–172. <https://doi.org/10.3390/polym3010150>
- Wang C, Zhang M, Xia K, Gong X, Wang H, Yin Z, Guan B, Zhang Y (2017) Intrinsically stretchable and conductive textile by a scalable process for elastic wearable electronics. *ACS Appl Mater Interfaces* 9:13331–13338. <https://doi.org/10.1021/acsami.7b02985>
- Webbe RE, Creton C, Brown HR, Gong JP (2007) Large strain hysteresis and Mullins effect of tough double-network hydrogels. *Macromolecules* 40:2919–2927. <https://doi.org/10.1021/ma062924y>
- Wei Y, Chen S, Lin Y, Yuan X, Liu L (2016) Silver nanowires coated on cotton for flexible pressure sensors. *J Mater Chem C* 4:935–943. <https://doi.org/10.1039/c5tc03419a>
- Wu H, Huang Y, Xu F, Duan Y, Yin Z (2016) Energy harvesters for wearable and stretchable electronics: from flexibility to stretchability. *Adv Mater* 28:9881–9919. <https://doi.org/10.1002/adma.201602251>
- Xiao Y, Cao Y, Xin B, Liu Y, Chen Z, Lin L, Sun Y (2018a) Fabrication and characterization of electrospun cellulose/polyacrylonitrile nanofibers with Cu(II) ions. *Cellulose* 25:2955–2963. <https://doi.org/10.1007/s10570-018-1784-5>
- Xiao Y, Xin B, Chen Z, Lin L, Liu Y, Hu Z (2018b) Enhanced thermal properties of graphene-based poly(vinyl chloride) composites. *J Ind Text.* <https://doi.org/10.1177/1528083718760805>
- Xu Q-b, Xie L, Diao H, Li F, Zhang Y, Fu F, Liu X (2017) Antibacterial cotton fabric with enhanced durability prepared using silver nanoparticles and carboxymethyl chitosan. *Carbohydr Polym* 177:187–193. <https://doi.org/10.1016/j.carbpol.2017.08.129>
- Yetisen AK, Qu H, Manbachi A, Butt H, Dokmeci MR, Hinesroza JP, Skorobogatyi M, Khademhosseini A, Yun SH (2016) Nanotechnology in Textiles. *ACS Nano* 10:3042–3068. <https://doi.org/10.1021/acsnano.5b08176>
- Zeng W, Shu L, Li Q, Chen S, Wang F, X-m Tao (2014) Fiber-based wearable electronics: a review of materials, fabrication, devices, and applications. *Adv Mater* 26:5310–5336. <https://doi.org/10.1002/adma.201400633>
- Zhang L, Baima M, Andrew TL (2017a) Transforming commercial textiles and threads into sewable and weavable electric heaters. *ACS Appl Mater Interfaces* 9:32299–32307. <https://doi.org/10.1021/acsami.7b10514>
- Zhang X-q, Huang X-x, Zhang X-d, Xia L-z, Zhong B, Zhang T, Wen G (2017b) Cotton/rGO/carbon-coated SnO₂ nanoparticle-composites as superior anode for lithium ion battery. *Mater Des* 114:234–242. <https://doi.org/10.1016/j.matdes.2016.11.081>
- Zhang Z-h, Jiang G-d, Wu Y, Kong F, Huang J (2018) Surface functional modification of ultrahigh molecular weight polyethylene fiber by atom transfer radical polymerization. *Appl Surf Sci* 427:410–415. <https://doi.org/10.1016/j.apsusc.2017.08.159>
- Zhu C-h, Li L-m, Wang J-h, Wu Y-p, Liu Y (2017) Three-dimensional highly conductive silver nanowires sponges based on cotton-templated porous structures for stretchable conductors. *RSC Adv* 7:51–57. <https://doi.org/10.1039/c6ra25260e>